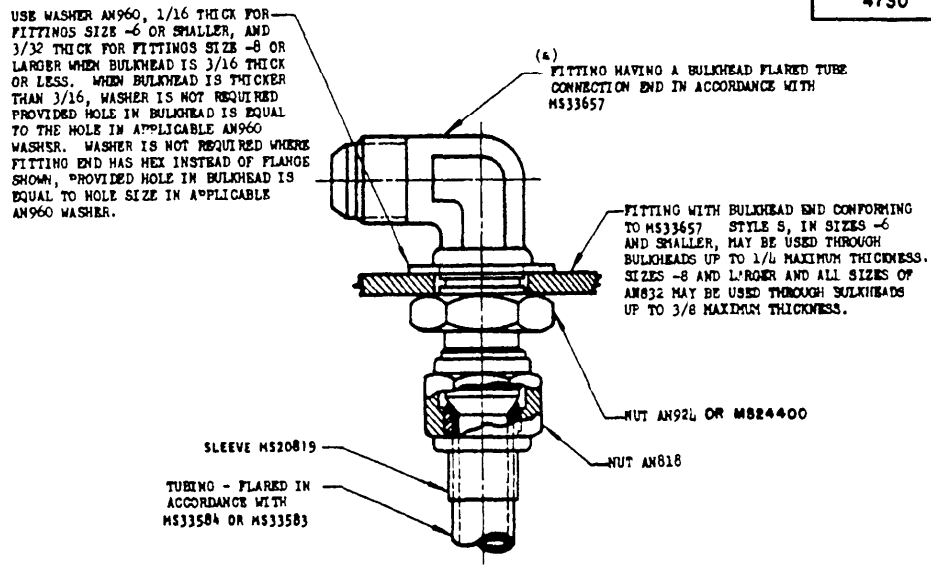


BULKHEAD AND FLARED TUBE INSTALLATION

TABLE I

DASH NOS. REF	TUBING OD INCHES	WRENCH TORQUE FOR TIGHTENING AN818 NUT (POUND INCH)						HOSE ASSEMBLIES (POUND INCH)	
		ALUMINUM-ALLOY TUBING		STEEL TUBING		ALUMINUM-ALLOY TUBING (FLARE MS33583) FOR USE ON OXYGEN LINES ONLY		MIL-H-8790 MIL-H-8795 MIL-H-25579	MIL-H-26666 MIL-H-38360 MIL-H-38390
		MINIMUM	MAXIMUM	MINIMUM	MAXIMUM	MINIMUM	MAXIMUM	MINIMUM	MAXIMUM
-2	1/8	20	30	75	85	---	---	SEE NOTES 1] AND 2] BELOW	SEE NOTES 1] AND 2] BELOW
-3	3/16	25	35	95	105	---	---		
-4	1/4	50	65	135	150	---	---		
-5	5/16	70	90	170	200	100	125		
-6	3/8	110	130	270	300	200	250		
-8	1/2	230	260	450	500	300	400		
-10	5/8	330	360	650	700	---	---		
-12	3/4	460	500	900	1000	---	---		
-16	1	500	700	1200	1400	---	---		
-20	1-1/4	800	900	1520	1680	---	---		
-24	1-1/2	800	900	1900	2100	---	---		
-28	1-3/4	---	---	---	---	---	---		
-32	2	1800	2000	2660	2940	---	---		

1] WHEN THE HOSE FITTING (NIPPLE AND NUT) IS ALUMINUM, THE MIN/MAX VALUES FOR ALUMINUM ALLOY TUBING SHALL APPLY.

2] WHEN THE HOSE FITTING (NIPPLE AND NUT) IS STEEL, THE MIN/MAX VALUES FOR STEEL TUBING SHALL APPLY.

INSTALLATION OF FITTING HAVING THE CONNECTION END IN ACCORDANCE WITH MS33656 IS SIMILAR, EXCEPT THAT THE END IS NOT USED THROUGH BULKHEADS. WRENCH TORQUES FOR ASSEMBLING THE TUBING ARE THE SAME.

NOMINAL USE: AIRFRAME FLUID CONNECTIONS

THIS IS DESIGN STANDARD AND SHALL NOT BE USED AS A PART NUMBER.

APPROVED 17 Aug 71 REVISED

User activities: Army - EL, MI
Review activities: Army - AV, MC, MU
This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. Air Force - 11
Other Coat
Navy - AS
Army - AV

TITLE
FITTINGS -- INSTALLATION OF FLARED TUBE, STRAIGHT THREADED CONNECTORS, DESIGN STANDARD FOR

MILITARY STANDARD
MS21344

PROCUREMENT SPECIFICATION

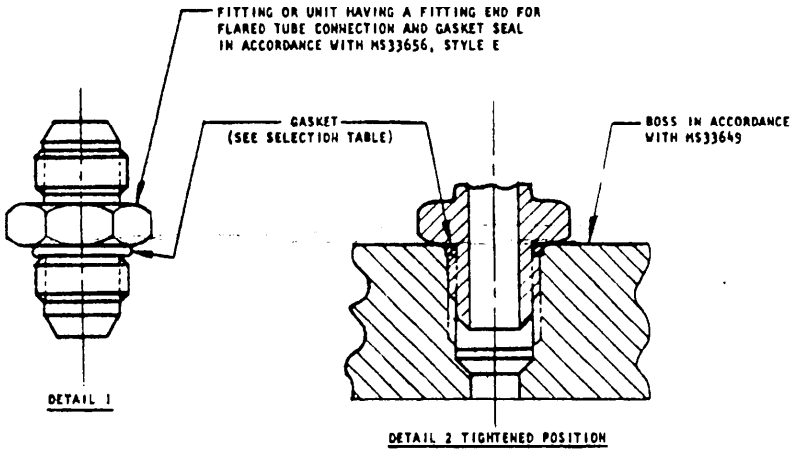
SUPERSEDES: AND10064

SHEET 1 OF 3

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NON-POSITIONING TYPE FITTING INSTALLATION SUITABLE FOR NOMINAL OPERATING PRESSURES UP TO AND INCLUDING 3000 PSI

PROCEDURE FOR INSTALLATION OF MS33656 FITTING END, STYLE E.

1. LUBRICATE THE GASKET IN APPROPRIATE LIQUID (SEE TABLE II).
2. INSTALL GASKET ON THE FITTING AS SHOWN IN DETAIL 1.
3. SCREW THE FITTING ASSEMBLY INTO THE BOSS UNTIL IT BOTTOMS TIGHTLY ON THE BOSS AS SHOWN IN DETAIL 2.

TABLE II

GASKET SELECTION TABLE

APPLICATION	MS NO.	APPROPRIATE LUBRICATING LIQUID
HYDRAULIC	MS28778	MIL-H-5606 OR PETROLATUM
ENGINE OIL	MS28778	ENGINE OIL
FUEL	MS29512	APPLICABLE FUEL
PNEUMATIC	MS28778	MIL-P-4343

User activities: Army - EL, MI
Review activities: Army - AV, WC, MU
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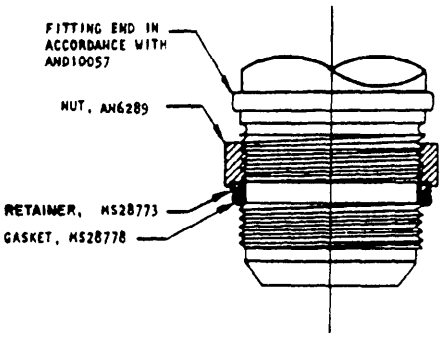
P.A. Air Force - 11 Other Cust Navy - AS Army - AV	TITLE FITTING--INSTALLATION OF FLARED TUBE, STRAIGHT THEADED CONNECTOR, DESIGN STANDARD FOR	MILITARY STANDARD MS21344
PROCUREMENT SPECIFICATION	SUPERSEDES: AND10064	SHEET 2 OF

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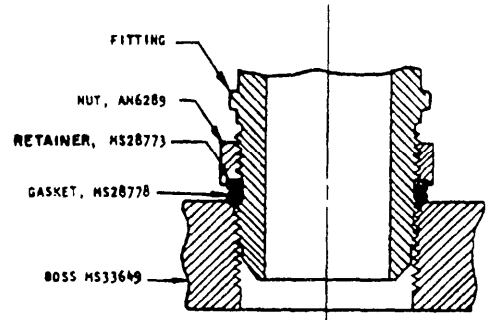
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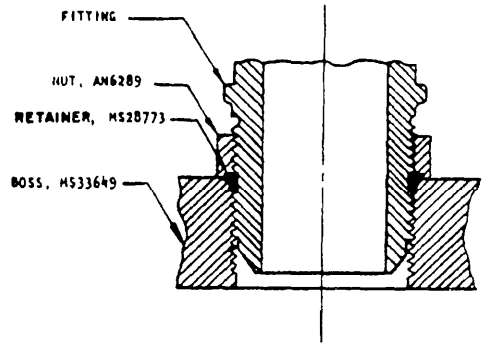
COAT MALE THREADS OF FITTING, RING, MS28773, AND GASKET, MS28778 SPARINGLY WITH PETROLATUM, SPECIFICATION AN-P-51 (VASELINE), OR HYDRAULIC FLUID AND ASSEMBLE AS SHOWN. WORK THE RETAINER MS28773 INTO THE COUNTERBORE OF THE NUT, AN6289; THEN TURN THE NUT, AN6289, DOWN UNTIL THE GASKET, MS28778, IS PUSHED FIRMLY AGAINST THE LOWER THREADED SECTION OF THE FITTING.

STEP 1



INSTALL THE FITTING INTO THE BOSS, MS33649, AND, AT THE SAME TIME, KEEP THE NUT, AN6289, TURNING WITH THE FITTING UNTIL THE GASKET, MS28778 CONTACTS THE BOSS, MS33649. THIS POINT CAN BE DETERMINED BY A SUDDEN INCREASE IN TORQUE. WITH THE FITTING IN THIS POSITION, PUT A WRENCH ON THE NUT, AN6289, TO PREVENT ITS TURNING AND, AT THE SAME TIME, TURN THE FITTING IN 1-1/2 TURNS. POSITION FITTING BY TURNING IN NOT MORE THAN ONE ADDITIONAL TURN.

STEP 2



HOLD FITTING AND TURN NUT, AN6289, DOWN TIGHTLY AGAINST BOSS, MS33649. SLIGHT EXTRUSION OF THE RING, MS28773 IS NOT DETRIMENTAL.

STEP 3

POSITIONING TYPE UNIVERSAL FITTING INSTALLATION SUITABLE FOR NOMINAL OPERATING PRESSURES UP TO AND INCLUDING 3,000 PSI

APPROVED 17 Aug 71 REVISED

User activities: Army - EL, MI

Review activities: Army - AV, WC, MU

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P.A. Air Force - 11 Other Cust Navy - AS Army - AV	TITLE FITTINGS - - INSTALLATION OF FLARED TUBE, STRAIGHT THREADED CONNECTORS, DESIGN STANDARD FOR	MILITARY STANDARD MS21344
PROCUREMENT SPECIFICATION	SUPERSEDES: AND10084	SHEET 3 OF

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